

2's complement:

$$1000 : -8$$

$$1001 : -7$$

$$1010 : -6$$

$$1011 : -5$$

$$1100 : -4$$

$$1101 : -3$$

$$1110 : -2$$

$$1111 : -1$$

In signed magnitude and 1's comp. rep. we have two diff. rep. for +0 or -0 so there is comparison betⁿ them and we need extra bits (space) for it.

while in 2's complement range is betⁿ $+(2^{n-1}-1)$ to -2^{n-1} [-8 to +7].
(+7) (-8)

→ In signed Magnitude representation, addition of two +ve numbers

$$0110 \rightarrow +6$$

$$0100 \rightarrow +4$$

$$\hline 1010$$

negative

In two -ve numbers,

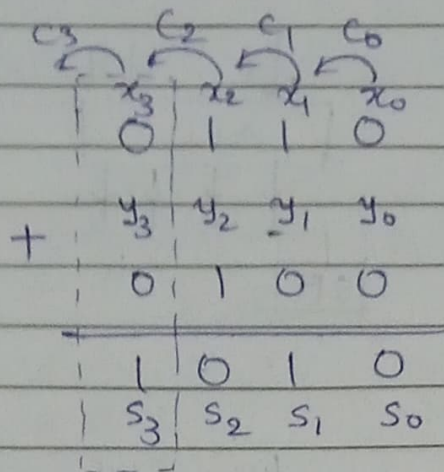
$$1010 \rightarrow -2$$

$$1011 \rightarrow -3$$

$$\hline 10101$$

positive

→ we have two numbers $X(x_3x_2x_1x_0)$ & $Y(y_3y_2y_1y_0)$ and carry is $c_3c_2c_1c_0$, then sum is $s_3s_2s_1s_0$.

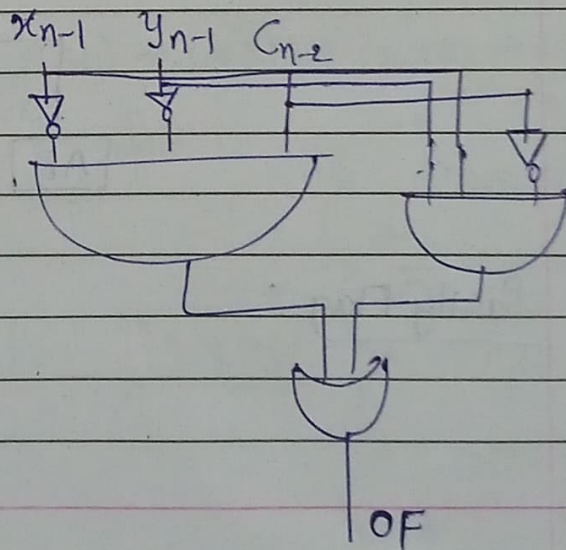


	x_3	y_3	c_2	s_3	c_3	Overflow Flag (OF)
	0	0	0	0	0	0
(Two +ve)	0	0	1	(-ve) 1	0	1
	0	1	0	1	0	0
	0	1	1	0	1	0
	1	0	0	1	0	0
	1	0	1	0	1	0
(Two -ve)	1	1	0	(+ve) 0	1	1
	1	1	1	1	1	0

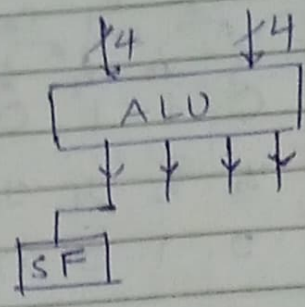
$$OF = x_3' y_3' c_2 + x_3 y_3 c_2' \quad (4 \text{ bit no.})$$

$$OF = x_{n-1}' y_{n-1}' c_{n-2} + x_{n-1} y_{n-1} c_{n-2}' \quad (n \text{-bit no.})$$

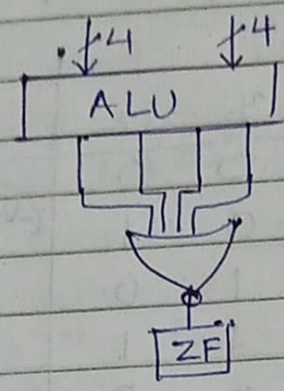
Logic Diagram:



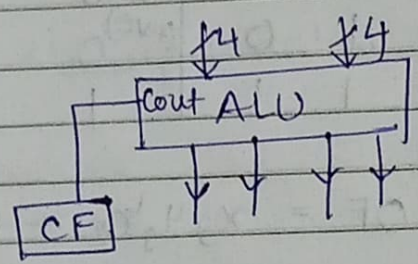
→ Sign Flag :



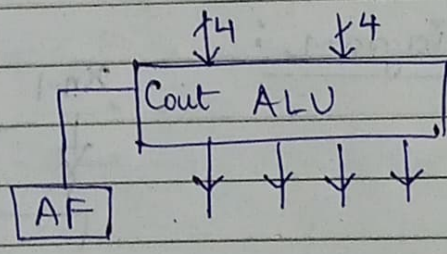
→ Zero flag :



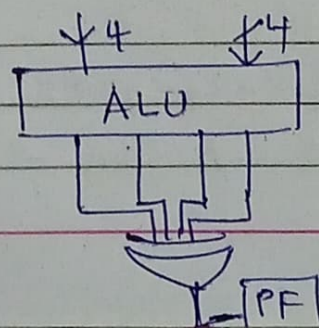
→ Carry flag :



→ Auxiliary flag :

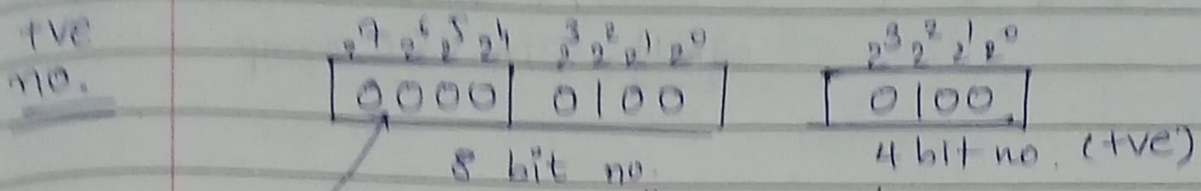


→ Parity Flag :



(even parity)

★ Unsigned Numbers

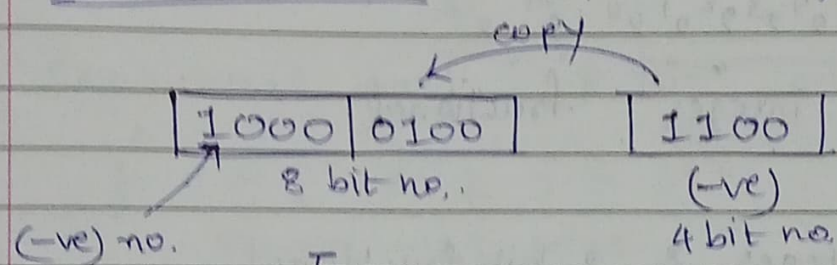


Since it is unsigned no. this should be 0.

want to copy

- this is same for signed rep, 1's comp & 2's comp.

→ For (-ve) no. : (Signed magnitude rep.)



[magnitude won't be change]

extra 2's

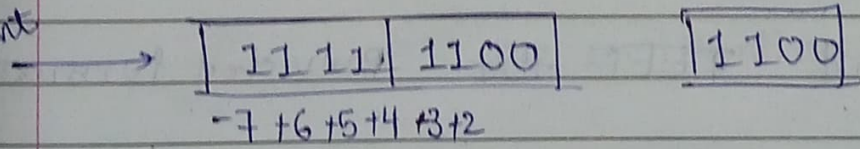
1 1 0 0

weight - 8 4 2 1 = $(-8 \times 1) + (4 \times 1) + (2 \times 0) + (1 \times 0)$

= -4 no.

In 2's complement :

1's complement



00000001	1
+	1
00000100	
(-4)	

0011	3
+	1
0100	
(-4)	

ex. 2

$$\begin{array}{cccc} 1 & 1 & 1 & 1 \\ -8 & 4 & 2 & 1 \end{array} = (-8 \times 1) + (4 \times 1) + (2 \times 1) + (1 \times 1) = -1 \text{ in 2's comp.}$$

$$\begin{array}{r} 1111 \\ 0000 \\ + \quad 1 \\ \hline 0001 \end{array} \quad \begin{array}{l} \text{2's comp.} \end{array}$$

$$\begin{array}{r} 1 \leftarrow \text{no.} \\ 0001 \\ 1110 \\ + \quad 1 \\ \hline 1111 \end{array} \quad \begin{array}{l} \text{1's comp.} \end{array}$$

ex. Binary Representation 127.25 (fixed point rep.)

$$\begin{array}{cccccccc} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 \\ 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 & 2^{-1} & 2^{-2} \end{array}$$

integer fractional

00 - 0.00

01 - 0.25 5 bit integer & 2 bit fractional

10 - 0.50 max. value can be represented = 31.75

11 - 0.75

↑
can only
be represented

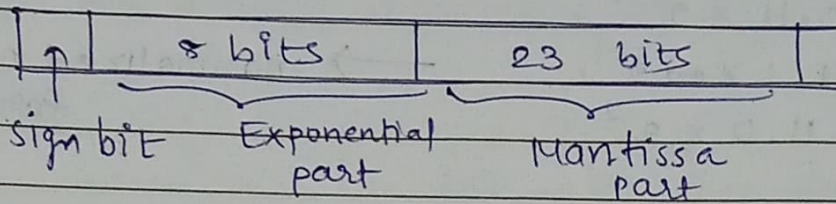
31.705 } — not possible to represent
0.02 IF we want to rep. then
we have to increase the
fractional bit.

ex In floating point representation

Standard IEEE 754 provided format,



32 bit



Any no. we want to represent, $\rightarrow 9^E \cdot (1.M)$

$$0.75 \rightarrow 3 \times 2^{-2} \quad \begin{matrix} \uparrow \\ \text{Mantissa} \end{matrix} \quad \begin{matrix} \nwarrow \\ (E) \end{matrix}$$

In general, $M \times B^E \rightarrow \text{EXP}$
 $\begin{matrix} \uparrow & \nwarrow \\ \text{Mantissa} & \text{Base} \end{matrix}$

Exponential part, 8 bit can be represented

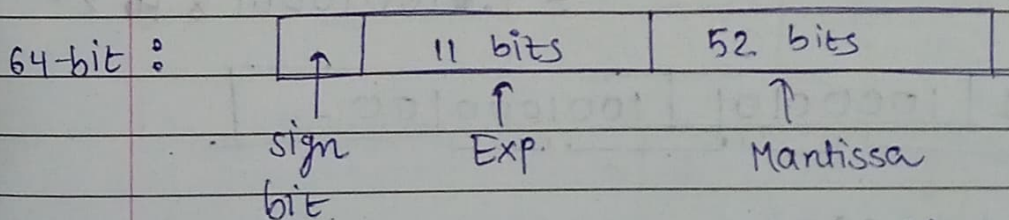
$$2^{8-1} - 1 = 2^7 - 1 = 127$$

Excess -127 code.

$$\begin{aligned} 0 &\rightarrow 01111111 \\ 1 &\rightarrow 1 + 127 = 128 \\ &\quad 10000000 \end{aligned}$$

32-bit = single precision

64-bit = double precision



$$2^{11-1} - 1 = 2^{10} - 1 = 1023 \quad (\text{Excess-1023 code})$$

Binary: 0.011×2^1
 0.11×2^0
 1.1×2^{-1}
 11.0×2^{-2}

→ normalized number form.

Represent $+1.1 \times 2^{-1}$ in 82-bit

sign bit = 0

exp. part = number

* 12.3

$$12.4 + (-1) = 12.6$$

0	0	1	1	1	1	1	0	1	000	---
---	---	---	---	---	---	---	---	---	-----	-----

Hex : 9F400000

eg. 2.5×100.25 in 32-bit format

(b) $2.5 \rightarrow$ Binary $10.1 = 1.01 \times 2^1$

O	100000000	0.000
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Hex : 40200000

(ii) $-100.25 \rightarrow$ Binary 10100100.01
 $= 1.010010001 \times 10^2$

1	1000000	1001000000
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Hex: C2C89000



Min exp. value : $1 - 127 = -126$

Max. exp. value : $254 - 127 = 127$

①

when exp. is $000 \dots 0$

Mantissa is $00 \dots 0$

$$N = (-1)$$

In general,

$$N = (-1)^{\text{sign}} \cdot 2^{E-127} \cdot (1.M)$$

②

when exp is $111 \dots 1$

Mantissa $\neq 0$

$$N = \text{NaN}$$

③

when exp is $111 \dots 1$

Mantissa = 0

$$N = -\infty \text{ to } +\infty$$

④

sign bit = 0, exp. = 1, Mantissa = 0

$$N = -2^{-126} \text{ (Minimum Number)}$$

$1 - 127$

↓

$$[2^{-126} (1.M)]$$

⑤

Maximum number = $2^{127} (1.111111 \dots 1)$

↑
 2^{-1}

↑
 2^{-23}

$$= 2^{127} (1 + 0.1111 \dots 1)$$

$$= 2^{127} \left(1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{23}} \right)$$

$$= 2^{127} \left[\frac{1 - (1/2)^{24}}{1 - (1/2)} \right]$$

$$= 2^{127} \left(2 - \left(\frac{1}{2} \right)^{23} \right)$$

Denormalized Form

$$\text{sign} = x$$

$$\text{exp.} = 00000000$$

$$\text{Mantissa} = 0.M$$

$$N = (-1)^{\text{sign}} \cdot 2^{E-126} (0.M)$$

$$\begin{aligned} \text{Minimum no.} &= (-1)^{\text{sign}} \cdot 2^{-126} (0.0000 \dots 0) \\ &= (-1)^{\text{sign}} \cdot 2^{-126} \cdot 2^{-23} \\ &= 2^{-149} \end{aligned}$$

eg.

C2C88000

D3D88000

1 1 0 1 0 0 1 1 1 0 1 1 0 0 0 1 0 0 0 0 0 0

$$\text{sign bit} = 1$$

$$\text{exp bit} = 10100111 \rightarrow \begin{matrix} 128 \\ + \\ 32 \\ + \\ 7 \end{matrix} \Bigg] 167$$

$$\text{Mantissa} = 101100010 \dots$$

$$\text{Minimum no.} = (-1)^1 \cdot 2^{167-126} (1.101100010 \dots)$$

$$= -2^{141}$$

$$= (-1)^1 \cdot 2^{40} (1 + 0.6914)$$

$$= -1.0995 \times 10^{12} \times 1.6914$$

$$= \boxed{-1.85969 \times 10^{12}}$$